

Programmability: T1 2D fabric vs SCAMP-5 pixel-processor array

SCAMP-5 has one analog processor per pixel but its ONLY links are the 4 nearest neighbours (N/E/W/S), run in lock-step (SIMD). Non-local moves must walk cell-by-cell; Non-uniform ops (each pixel a different distance from the target) need many shift rounds. T1 reaches any lane directly (gather / slide / H-V transpose) and reduces in hardware.

SCAMP-5 PE — 4 neighbours only

N

W

E

S

data moves 1 cell per instruction

T1 lane — gather reaches ANY lane

SRC

DST

any source → any destination in same lane group in ONE instruction

Task A · move one pixel: top-left → top-right → bottom-right

T1 fabric

start

→ vrgather (H) →

top-right

→ vrgather (V) →

bottom-right

✓ 2 instructions

SCAMP-5

start

→ news E →

news E

→ news E →

news E

→ news S →

news S

→ news S →

bottom-right

× 6 neighbour-shifts (+ mask)

Task B · sum each row into the left column (vredsum) — an non-uniform move: every pixel is a different distance from the target

T1 fabric

3

1

4

2

1

5

9

2

6

5

3

5

2

7

1

8

→ vredsum.vs →

10

.

.

.

17

.

.

.

19

.

.

.

18

.

.

.

row-sums in col 0

✓ 1 instruction (every distance at once)

SCAMP-5

3

1

4

2

1

5

9

2

6

5

3

5

2

7

1

8

→ shift 1 + add →

4

5

6

2

6

14

11

2

11

8

8

5

9

8

9

8

after round 1

→ shift 2 + add →

10

7

6

2

17

16

11

2

19

13

8

5

18

16

9

8

row-sums in col 0

× log₂W rounds, W−1 shifts (uniform only)

Summary

Aspect	SCAMP-5	T1 2D fabric
Inter-PE connectivity	4 nearest neighbours (N/E/W/S)	any pixel in same row/col → any pixel in same row/col (gather) + H/V transpose
Relocate a pixel by (dx,dy)	dx + dy sequential shifts	1 gather (vrgather)
Sum each row	log ₂ W shift-add rounds, all PEs shift together	1 vredsum (hardware reduction)
Control granularity	SIMD: every pixel moves the SAME way without masking	High level lane process control, uniform & non-uniform, regular & irregular ops